



GOA COLLEGE OF ENGINEERING

"BHAUSAHEB BANDODKAR TECHNICAL EDUCATION COMPLEX"
FARMAGUDI, PONDA- GOA - INDIA

MATHEMATICS-III

UNIT 1: MATRICES

LECTURE 8

**DEVELOPED BY MATHEMATICS FACULTY
DEPARTMENT OF SCIENCE & HUMANITIES
GOA COLLEGE OF ENGINEERING**



MATRICES LECTURE 8

TOPICS TO BE COVERED:

- Cayley-Hamilton Theorem and Applications
- Diagonalization of a matrix



Cayley-Hamilton Theorem

Theorem: Every square matrix satisfies its own Characteristic equation.

Or

If A is square matrix of order n , and $|A - \lambda I| = 0$ i.e.

$(-1)^n \lambda^n + k_1 \lambda^{n-1} + \dots + k_{n-1} \lambda + k_n = 0$ is characteristic equation of A , then

$$(-1)^n A^n + k_1 A^{n-1} + \dots + k_{n-1} A + k_n I = 0$$

Proof: Consider the matrix $(A - \lambda I)$ and $P = \text{adj}(A - \lambda I)$.

Since the elements of $A - \lambda I$ are at most of the first degree in λ , the elements of $\text{adj}(A - \lambda I)$ are at most degree $(n - 1)$ in λ .

Thus, $\text{adj}(A - \lambda I)$ may be written as a matrix polynomial in λ , given by

$$\text{adj}(A - \lambda I) = P = P_0 \lambda^{n-1} + P_1 \lambda^{n-2} + \dots + P_{n-1}$$

where $P_0, P_1 \dots P_{n-1}$ are $n \times n$ matrices.



Cayley-Hamilton Theorem

We know that $(A - \lambda I) \cdot \text{adj}(A - \lambda I) = |(A - \lambda I)|I$

$$(A - \lambda I)(P_0\lambda^{n-1} + P_1\lambda^{n-2} + \dots + P_{n-1}) = [(-1)^n\lambda^n + k_1\lambda^{n-1} + \dots + k_{n-1}\lambda + k_n].I$$

Equating Coefficient of like powers of λ on both the sides we get

$$-P_0 = (-1)^n.I$$

$$AP_0 - P_1 = k_1.I$$

$$AP_1 - P_2 = k_2.I$$

$$AP_2 - P_3 = k_3.I$$

$$\vdots \quad \quad \quad \vdots$$

$$AP_{n-1} = k_n.I$$

On multiplying the equations by $A^n, A^{n-1}, \dots, I$ respectively and adding, we obtain

$$0 = (-1)^n A^n + k_1 A^{n-1} + \dots + k_{n-1} A + k_n I$$

$$\text{Thus } (-1)^n A^n + k_1 A^{n-1} + \dots + k_{n-1} A + k_n I = \mathbf{0}.$$



Application

- To find A^{-1} :

By Cayley Hamilton Theorem,

$$(-1)^n A^n + k_1 A^{n-1} + \dots + k_{n-1} A + k_n I = \mathbf{0} .$$

Multiply this equation by A^{-1} ,

$$[(-1)^n A^n + k_1 A^{n-1} + \dots + k_{n-1} A + k_n I] \cdot A^{-1} = \mathbf{0} .$$

$$k_n A^{-1} = -[(-1)^n A^{n-1} + k_1 A^{n-2} + \dots + k_{n-1} I]$$

$$A^{-1} = \frac{-1}{k_n} [(-1)^n A^{n-1} + k_1 A^{n-2} + \dots + k_{n-1} I]$$



Examples

1. Verify Cayley hamilton theorem for the matrix

$$A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} \text{ and hence find } A^{-1}.$$

Solution: Characteristic equation is $|A - \lambda I| = 0$

$$\lambda^3 - (2 + 2 + 2)\lambda^2 + \left\{ \begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} + \begin{vmatrix} 2 & -1 \\ 1 & 2 \end{vmatrix} + \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} \right\} \lambda - \begin{vmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{vmatrix} = 0$$

$$-\lambda^3 + 6\lambda^2 - 9\lambda + 4 = 0$$

$$\text{Or } \lambda^3 - 6\lambda^2 + 9\lambda - 4 = 0$$

$$\text{To prove } A^3 - 6A^2 + 9A - 4I = 0$$



Examples

Verification:

$$A^2 = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix} = \begin{bmatrix} 22 & -21 & 21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{bmatrix}$$

$$A^3 - 6A^2 + 9A - 4I =$$

$$= \begin{bmatrix} 22 & -21 & 21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{bmatrix} - 6 \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix} + 9 \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} - 4 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 22 - 36 + 18 - 4 & -21 + 30 - 9 - 0 & 21 - 30 + 9 - 0 \\ -21 + 30 - 9 - 0 & 22 - 36 + 18 - 4 & -21 + 30 - 9 - 0 \\ 21 - 30 + 9 - 0 & -21 + 30 - 9 - 0 & 22 - 36 + 18 - 4 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$



Examples

- **Inverse of a matrix:**

$$A^3 - 6A^2 + 9A - 4I = 0$$

On multiplying by A^{-1} , we get

$$A^2 - 6A + 9I - 4A^{-1} = 0$$

$$A^{-1} = \frac{1}{4}(A^2 - 6A + 9I)$$

$$A^{-1} = \frac{1}{4} \left\{ \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix} - 6 \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} + 9 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\}$$

$$A^{-1} = \frac{1}{4} \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix}$$



Examples

2. Verify Cayley hamilton theorem for the matrix

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$$

and hence find the matrix represented by

$$A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$$

Solution: Characteristic equation is $|A - \lambda I| = 0$

$$\lambda^3 - (2 + 1 + 2)\lambda^2 + \left\{ \begin{vmatrix} 2 & 1 \\ 0 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 0 \\ 1 & 2 \end{vmatrix} + \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} \right\} \lambda - \begin{vmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{vmatrix} = 0$$

$$\lambda^3 - 5\lambda^2 + 7\lambda - 3 = 0$$

Verify that $A^3 - 5A^2 + 7A - 3I = 0$



Examples

$$\begin{aligned} &\text{Consider } A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I \\ &= A^5 (A^3 - 5A^2 + 7A - 3I) + A^4 - 5A^3 + 7A^2 - 3A + A^2 + A + I \\ &= A^5 \cdot 0 + A(A^3 - 5A^2 + 7A - 3I) + A^2 + A + I \\ &= A \cdot 0 + A^2 + A + I \\ &= A^2 + A + I \end{aligned}$$

$$= \begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix} + \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 8 & 5 & 5 \\ 0 & 3 & 0 \\ 5 & 5 & 8 \end{bmatrix}$$



Similar matrices

Similar Matrices: Two matrices A and B are said to be similar, if there exists a non-singular matrix P such that $B = P^{-1}AP$ and is denoted by $A \sim B$.

Theorem: Two similar matrices have the same eigenvalues.

Proof: Let A and B be two similar matrices.

Therefore, there exists a non-singular matrix P such that $B = P^{-1}AP$

$$B - \lambda I = P^{-1}AP - \lambda I$$

$$B - \lambda I = P^{-1}AP - P^{-1}\lambda IP$$

$$B - \lambda I = P^{-1}(A - \lambda I)P$$

$$|B - \lambda I| = |P^{-1}(A - \lambda I)P|$$

$$= |P^{-1}| |(A - \lambda I)| |P|$$

$$= |(A - \lambda I)|$$

Since A and B have the same characteristic polynomial, they have same eigenvalues.



Diagonalization of a matrix

Definition: Let A be a square matrix. The process of finding a non-singular matrix P such that $P^{-1}AP = D$ where D is a diagonal matrix, is called diagonalization of matrix A .

Theorem: If a square matrix A of order n has n linearly independent eigenvectors, then a matrix P can be found such that $P^{-1}AP$ is a diagonal matrix.

Proof: We will prove this result for $n=3$.

Let A be a square matrix of order 3.

Let $\lambda_1, \lambda_2, \lambda_3$ be its eigenvalues and $X_1 = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}$, $X_2 = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$, $X_3 = \begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix}$ be its corresponding eigenvectors which are linearly independent..

Then $A X_1 = \lambda_1 X_1$, $A X_2 = \lambda_2 X_2$ and $A X_3 = \lambda_3 X_3$

Let $P = [X_1 \quad X_2 \quad X_3] = \begin{bmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{bmatrix}$



Diagonalization of a matrix

$$\begin{aligned} AP = A[X_1 \quad X_2 \quad X_3] &= [AX_1 \quad AX_2 \quad AX_3] = \begin{bmatrix} \lambda_1 x_1 & \lambda_2 x_2 & \lambda_3 x_3 \\ \lambda_1 y_1 & \lambda_2 y_2 & \lambda_3 y_3 \\ \lambda_1 z_1 & \lambda_2 z_2 & \lambda_3 z_3 \end{bmatrix} \\ &= \begin{bmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} = PD \end{aligned}$$

$$P^{-1}AP = D$$

- The matrix P which diagonalizes A is called the **modal matrix** of A and is obtained by grouping the eigenvectors of A into a square matrix .
- The resultant diagonal matrix D is called the **spectral matrix** and has the eigenvalues of A as its elements.



Diagonalization of a matrix

To find A^n :

$$P^{-1}AP = D$$

$$A = PDP^{-1}$$

$$A^2 = (PDP^{-1})(PDP^{-1}) = PD^2P^{-1}$$

$$A^3 = (PDP^{-1})(PDP^{-1})(PDP^{-1}) = PD^3P^{-1}$$

$$\vdots \quad \quad \quad \vdots$$

$$A^n = PD^nP^{-1}$$

$$A^n = P \begin{bmatrix} \lambda_1^n & 0 & \cdots & 0 \\ 0 & \lambda_2^n & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n^n \end{bmatrix} P^{-1}$$



Examples

1. Diagonalize, if possible, the matrix $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$. Hence find A^4 .

Solution: The Characteristic equation of A is $|A - \lambda I| = 0$.

$$\lambda^3 - 7\lambda^2 + 36 = 0$$

$\lambda_1 = -2, \lambda_2 = 3$ and $\lambda_3 = 6$ are the **Eigenvalues** of A and $X_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$, $X_2 = \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}$ and

$X_3 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$ are the corresponding eigenvectors.

Since $\lambda_1 = -2, \lambda_2 = 3$ and $\lambda_3 = 6$ are distinct, the corresponding eigenvectors X_1, X_2 and X_3 are linearly independent.



Examples

$$\text{Now } P = \begin{bmatrix} -1 & -1 & 1 \\ 0 & 1 & 2 \\ 1 & -1 & 1 \end{bmatrix}$$

$$P^{-1} = -\frac{1}{6} \begin{bmatrix} 3 & 0 & -3 \\ 2 & -2 & 2 \\ -1 & -2 & -1 \end{bmatrix}$$

$$P^{-1}AP = -\frac{1}{6} \begin{bmatrix} 3 & 0 & -3 \\ 2 & -2 & 2 \\ -1 & -2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix} \begin{bmatrix} -1 & -1 & 1 \\ 0 & 1 & 2 \\ 1 & -1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 6 \end{bmatrix} = D$$

$$A^4 = PD^4P^{-1} = \begin{bmatrix} -1 & -1 & 1 \\ 0 & 1 & 2 \\ 1 & -1 & 1 \end{bmatrix} \cdot \begin{bmatrix} (-2)^4 & 0 & 0 \\ 0 & 3^4 & 0 \\ 0 & 0 & 6^4 \end{bmatrix} \cdot -\frac{1}{6} \begin{bmatrix} 3 & 0 & -3 \\ 2 & -2 & 2 \\ -1 & -2 & -1 \end{bmatrix} = \begin{bmatrix} 251 & 405 & 235 \\ 405 & 891 & 405 \\ 235 & 405 & 251 \end{bmatrix}$$



Examples

2. Diagonalize, if possible, the matrix $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

Solution: The Characteristic equation of A is $|A - \lambda I| = 0$.

$$\lambda^3 - (0 + 0 + 0)\lambda^2 + \left\{ \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} + \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} + \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} \right\} \lambda - \begin{vmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{vmatrix} = 0$$
$$\lambda^3 - 3\lambda - 2 = 0$$

$\lambda_1 = -1, \lambda_2 = -1$ and $\lambda_3 = 2$ are the **Eigenvalues** of A

To find eigenvectors, consider $(A - \lambda I)X = 0$

$$\begin{bmatrix} -\lambda & 1 & 1 \\ 1 & -\lambda & 1 \\ 1 & 1 & -\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \dots\dots\dots (1)$$



Examples

1. Put $\lambda = -1$ in equation (1)

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x_1 + x_2 + x_3 = 0$$

$$x_1 + x_2 + x_3 = 0$$

$$x_1 + x_2 + x_3 = 0$$

Note: For the homogenous system $AX = 0$ in n unknowns, there are $(n-r)$ linearly independent solutions where r is the rank of A

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \xrightarrow[R_3 \rightarrow R_3 - R_1]{R_2 \rightarrow R_2 - R_1} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Since $\text{Rank}(A - \lambda I) = 1$, there are $(3-1)=2$ linearly independent solutions(eigenvectors).

Let $x_3 = k$ and $x_2 = l$ be the two free variables. $\therefore x_1 = -k - l$

(i) Taking $k = 1$ and $l = 0$ we get, $X_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$



Examples

(ii) Taking $k = 0$ and $l = 1$, we get, $X_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$

2. Put $\lambda = 2$ in equation (1)

$$\begin{bmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$
$$-2x_1 + x_2 + x_3 = 0$$

$$x_1 - 2x_2 + x_3 = 0$$

$$x_1 + x_2 - 2x_3 = 0$$

From first and second equation,

$$\frac{x_1}{\begin{vmatrix} 1 & 1 \\ -2 & 1 \end{vmatrix}} = \frac{x_2}{-\begin{vmatrix} -2 & 1 \\ 1 & 1 \end{vmatrix}} = \frac{x_3}{\begin{vmatrix} -2 & 1 \\ 1 & -2 \end{vmatrix}} \Rightarrow \frac{x_1}{3} = \frac{x_2}{3} = \frac{x_3}{3} \Rightarrow \frac{x_1}{1} = \frac{x_2}{1} = \frac{x_3}{1}$$

Therefore the Eigenvector corresponding to eigenvalue $\lambda_3 = 2$ is $X_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$



Examples

$$\text{Now } P = \begin{bmatrix} -1 & -1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

$$P^{-1} = -\frac{1}{3} \begin{bmatrix} 1 & 1 & -2 \\ 1 & -2 & 1 \\ -1 & -1 & -1 \end{bmatrix}$$

$$\begin{aligned} P^{-1}AP &= -\frac{1}{3} \begin{bmatrix} 1 & 1 & -2 \\ 1 & -2 & 1 \\ -1 & -1 & -1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} -1 & -1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{bmatrix} = D \end{aligned}$$



Examples

3. Diagonalize, if possible, the matrix $A = \begin{bmatrix} 2 & -2 & 2 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix}$

Solution: The Characteristic equation of A is $|A - \lambda I| = 0$.

$$\lambda^3 - (2 + 1 - 1)\lambda^2 + \left\{ \begin{vmatrix} 2 & -2 \\ 1 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 1 \\ 3 & -1 \end{vmatrix} + \begin{vmatrix} 2 & 2 \\ 1 & -1 \end{vmatrix} \right\} \lambda - \begin{vmatrix} 2 & -2 & 2 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{vmatrix} = 0$$

$$\lambda^3 - 2\lambda^2 - 4\lambda + 8 = 0$$

$\lambda_1 = -2, \lambda_2 = 2$ and $\lambda_3 = 2$ are the **Eigenvalues** of A

To find eigenvectors, consider $(A - \lambda I)X = 0$

$$\begin{bmatrix} 2 - \lambda & -2 & 2 \\ 1 & 1 - \lambda & 1 \\ 1 & 3 & -1 - \lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \dots\dots\dots (1)$$



Examples

1. Put $\lambda = -2$ in equation (1)

$$\begin{bmatrix} 4 & -2 & 2 \\ 1 & 3 & 1 \\ 1 & 3 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$4x_1 - 2x_2 + 2x_3 = 0$$

$$x_1 + 3x_2 + x_3 = 0$$

$$x_1 + 3x_2 + x_3 = 0$$

From first and second equation,

$$\frac{x_1}{\begin{vmatrix} -2 & 2 \\ 3 & 1 \end{vmatrix}} = \frac{x_2}{-\begin{vmatrix} 4 & 2 \\ 1 & 1 \end{vmatrix}} = \frac{x_3}{\begin{vmatrix} 4 & -2 \\ 1 & 3 \end{vmatrix}} \Rightarrow \frac{x_1}{-8} = \frac{x_2}{-2} = \frac{x_3}{14} \Rightarrow \frac{x_1}{-4} = \frac{x_2}{-1} = \frac{x_3}{7}$$

Therefore the Eigenvector corresponding to eigenvalue $\lambda_1 = -2$ is $X_1 = \begin{bmatrix} -4 \\ -1 \\ 7 \end{bmatrix}$



Examples

2. Put $\lambda = 2$ in equation (1)

$$\begin{bmatrix} 0 & -2 & 2 \\ 1 & -1 & 1 \\ 1 & 3 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-2x_2 + 2x_3 = 0$$

$$x_1 - x_2 + x_3 = 0$$

$$x_1 + 3x_2 - 3x_3 = 0$$

Since $\text{Rank}(\mathbf{A} - \lambda\mathbf{I}) = 2$, there is $(3 - 2) = 1$ linearly independent solution (eigenvector).

From first and second equation,

$$\frac{x_1}{\begin{vmatrix} -2 & 2 \\ -1 & 1 \end{vmatrix}} = \frac{x_2}{-\begin{vmatrix} 0 & 2 \\ 1 & 1 \end{vmatrix}} = \frac{x_3}{\begin{vmatrix} 0 & -2 \\ 1 & -1 \end{vmatrix}} \Rightarrow \frac{x_1}{0} = \frac{x_2}{2} = \frac{x_3}{2} \Rightarrow \frac{x_1}{0} = \frac{x_2}{1} = \frac{x_3}{1}$$

Therefore the Eigenvector corresponding to eigenvalue $\lambda_2 = 2$ is $\mathbf{X}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$

Since there are only two linearly independent eigenvectors, the matrix A is not diagonalisable.

$$\begin{aligned} \begin{bmatrix} 0 & -2 & 2 \\ 1 & -1 & 1 \\ 1 & 3 & -3 \end{bmatrix} &\xrightarrow{R_1 \leftrightarrow R_2} \sim \begin{bmatrix} 1 & -1 & 1 \\ 0 & -2 & 2 \\ 1 & 3 & -3 \end{bmatrix} \\ &\xrightarrow{R_3 \rightarrow R_3 - R_1} \sim \begin{bmatrix} 1 & -1 & 1 \\ 0 & -2 & 2 \\ 0 & 4 & -4 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 + 2R_2} \sim \begin{bmatrix} 1 & -1 & 1 \\ 0 & -2 & 2 \\ 0 & 0 & 0 \end{bmatrix} \\ &\xrightarrow{R_2 \rightarrow \frac{R_2}{-2}} \sim \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

THANK YOU!